Orqix change record

Wave A: connect to outside systems and wait for them

Features implemented, enhanced and fixed while closing the Camunda gaps of doc 29, with everything needed to apply the same changes to another copy of the code.

Item	Value
Date	6 November 2026
Repository branch	`claude/camunda-features-gap-a441b7`
Base commit	`d5b0b774` (Merge pull request #22, develop)
Final commit of this record	`1fc2d70b`
Tickets closed	CG-W01, CG-W03, CG-W04, CG-W08, CG-C01, CG-C02, CG-C08 / CG-I04, CG-C09, CG-E10 (plus the doc 29 plan itself)
Files changed	207 files across backend, compiler, contracts, frontend, SDK and docs
Database migrations	V210, V211, V220, V230, V240, V241, V242
Verification	Compiler 884 tests, application 1,239 tests (0 failures, 5 opt-in skips), frontend 1,415 tests, typecheck, lint and API contract check: all pass
Status	Committed locally; not pushed, no pull request yet

1. What changed, in one page

Before this work Orqix could not connect to a typical SaaS API (for example ServiceNow) without writing an external worker, could not keep checking another system until a job was done, and retried every failing call with one fixed delay. Wave A fixes that.

Ticket	Feature	Type	Build note
CG-W01	Keep checking until done: a call step repeats a read-only check on a schedule until a condition says done, failed or the deadline passes	New	S129
CG-W03	Polling safety net for message waits: resume on a webhook or a slow check, whichever comes first	New	S129
CG-W08	Waiting for: one projection, panel and API saying what every request is waiting on; operator Check now / Extend deadline / Stop waiting; Insights 'Waiting on outside systems'	New	S129
CG-C01	Connection definition v1: path templates, query parameters with safe encoders, headers, DELETE, nested/form/multipart/binary bodies, row lists, pagination, error mapping, 120 s long calls	Enhanced	S130
CG-C02	Lookup reconciliation (idempotentByLookup) and idempotentNatural: no duplicate records after an uncertain create, no manual reconcile	New	S130
CG-W04	Per-step retry policy: exponential backoff with jitter, Retry-After for 429/503, circuit breaker and request limit per connection binding	Enhanced	S131
CG-C09	Connection health in Settings: success rate, latency, 429s, token failures, circuit state, notices	New	S131
CG-C08 / CG-I04	Secret providers: HashiCorp Vault, AWS Secrets Manager, Azure Key Vault, Google Secret Manager, with a secret-access audit that never records values	New	S132
CG-E10	Worker business errors: an external worker can report a business error that takes the step's Failed route or the section's error route	New	S132
Integration fixes	Six merge overlaps combined; four defects found by the combined run fixed (section 8)	Fixed	This record

2. How to apply these changes in another copy

Option A (recommended): bring the commits across with Git

If the office copy shares history with this repository (it contains commit `d5b0b774`), fetch the branch and merge it. Everything, including conflict resolutions and fixes, comes with it.

```
git fetch <this-repo> claude/camunda-features-gap-a441b7
git checkout -b wave-a <your-base-branch>
git merge --no-ff FETCH_HEAD
# or apply as patches:
git format-patch d5b0b774..1fc2d70b -o wave-a-patches
git am --3way wave-a-patches/*.patch
```

When the office copy has drifted, cherry-pick in this order (the merges in between only combine branches; their resolutions are described in section 8 and must be redone by hand when cherry-picking):

#	Commit	Content
1	`b931c3a5`	docs: doc 29 gap analysis and plan, README link
2	`1411a52a`	compiler: Keep checking until done and message-wait safety net
3	`b19e7d09`	runtime: poll scheduler, safety-net checks, Waiting-for API
4	`c667c1c4`	studio: inspector, Waiting-for panel, simulator check sequences
5	`78865323`	integration: connection definition v1 and lookup reconciliation
6	`c07156bc`	step retry policy, Retry-After, circuit breaker, connection health
7	`003ea821`	secret providers with access audit, worker business errors
8	`9f80e10b`	fix: retryAfter in the paginated call result (merge fix)
9	`49541af6`	waits follow-ups after the merge (deferral, Retry-After, optional results, migration and repair coverage, purge, inspector, Insights, load test)

#	Commit	Content
10	`1fc2d70b`	fix: poll purge migration renumbered V212 to V242 and made rerunnable

Option B: re-implement by hand

Use sections 3 to 7 as the specification. Each section lists the contract fields, compiler rules, database migrations, API operations, permissions, configuration and the files that carry the change. The build notes `docs/build/S129` to `S132` in the repository hold the full detail and are part of the commits above.

Prerequisites and order

- JDK 21 (the build used `JAVA_HOME=C:/Users/karup/.jdks/jdk-21.0.12.1+1`); Maven wrapper `backend/mvnw` or the Gradle build; Node for the frontend.
- Database migrations run in version order: V210, V211, V220, V230, V240, V241, V242. They only add tables, columns, checks and grants; no data is rewritten. V242 is safe on a database that already ran an earlier V212.
- Regenerate frontend API types after taking `contracts/openapi.yaml`: `npm run api:generate`, then `npm run api:check`.
- Compiler version moves to 0.3.0 (one bump for Wave A). Existing workflows compile to the same BPMN unless they use the new options.
- Every new external secret store is off by default; nothing changes until it is enabled (section 6).

3. Keep checking until done, safety net and Waiting for (S129)

What it does

A **Call a system** step on a read-only operation gets **Keep checking until…** (`config.poll`). Orqix calls the operation, maps the answer and continues on `next` when `until` holds, takes `failed` when `failWhen` holds, and takes `timedOut` at the deadline or when the checks are used up. The repetition lives inside the step, so the workflow graph stays acyclic.

```
"poll": {
  "every": "PT15M", "backoff": { "multiplier": 2, "maxInterval": "PT2H" }, "jitterPercent": 10,
  "until":    { "fieldRef": "/steps/checkIncident/state", "operator": "in", "value": ["6", "7"] },
  "failWhen": { "fieldRef": "/steps/checkIncident/state", "operator": "eq", "value": "8" },
  "giveUpAfter": "P5D", "maxChecks": 200,
  "onCheckError": { "maxConsecutive": 5, "then": "needsAttention" },
  "finalCheckAtDeadline": true, "showWhileWaiting": ["state"], "optionalResults": []
}
```

How long it waits. The first check runs immediately; each interval is the previous one times the multiplier, capped at `maxInterval`, with ±jitter. Example: every 5 minutes, multiplier 2, cap 1 hour, give up after 2 days gives 52 checks (0:00, 0:05, 0:15, 0:35, 1:15, then hourly, plus a final check at the deadline). The request resumes at the first check after the other system finishes, so the worst-case delay is the current interval.

A **Wait for a message** step can add a safety net (`config.fallbackCheck`): the same call settings with a slow interval. It resumes on the message or on the check, whichever comes first, and never twice.

Compiler rules

Code	Rule
WF_POLL_READONLY	The operation must be readOnly (also for the safety net)
WF_POLL_INTERVAL	every PT10S to P1D (safety net PT1M to P1D)
WF_POLL_BACKOFF	maxInterval at least every, at most P1D; multiplier 1 to 4
WF_POLL_DEADLINE	giveUpAfter PT1M to P90D and at least every
WF_POLL_TOO_MANY_CHECKS	Worst case at most 1,000 checks without maxChecks
WF_POLL_ROUTES	Exactly one timedOut path; a failed path when failWhen is set
WF_POLL_CONDITION / WF_POLL_OVERLAP	Conditions name mapped results; until and failWhen can never both hold
WF_POLL_SHOW / WF_POLL_OPTIONAL	Shown and optional fields must be kept results
WF_POLL_COMBINATION	No polling on an undo call or a call that raises an error on failure
WF_POLL_FALLBACK_MAPPING	Safety-net results map to declared message payload fields

Runtime design

- The step compiles to a triggerable service task. Entering it inserts a `poll_wait` row (V210). `PollScheduler` (tick 2 s, worker and all-in-one pods only) claims up to 20 due rows with `FOR UPDATE SKIP LOCKED`, a lease and a fence, runs the check through `EffectDispatcher` (egress, credentials, timeout, journal), and records the outcome only if it still holds the fence.
- One journal entry per check (occurrence = check number), so a pod that dies after the call is re-claimed and reuses the recorded answer instead of calling again.
- After downtime a due row gets exactly one check, never one per missed interval. Suspended requests are not checked; cancelling a request closes its rows in the same transaction.
- No payloads at rest: the row keeps status, an outcome code, a digest and up to three shown values.
- Deferred calls (open circuit or rate limit) move the next check to the end of the wait without counting a check or an error; `Retry-After` sets the next check, capped at the deadline and maxInterval.
- Release migration of a polling request only when the step settings are identical (`POLL_CHANGED`); the repair scan reports poll and wait-state drift; ended rows are purged after 30 days (V242).

API operations (tag Waits)

Operation	Endpoint	Permission
listRunWaits	GET /workspaces/{ws}/runs/{runId}/waits	run.read (initiator or participant)
searchWaits	GET /workspaces/{ws}/waits?kind=&connectionKey=&waitingLongerThan=	run.read
checkWaitNow	POST .../runs/{runId}/waits/{waitId}/check-now	run.operate, reason, audited
extendWaitDeadline	POST .../waits/{waitId}/extend-deadline	run.operate, reason, audited
stopWaiting	POST .../waits/{waitId}/stop	run.operate, reason, audited
getInsightsOutsideWaits	GET /workspaces/{ws}/insights/outside-waits	run.read

Database and configuration

Item	Detail
V210__poll_wait.sql	poll_wait table (state, checks, next check, deadline, lease, fence, digest, shown values)
V211__wait_state.sql	wait_state projection for every wait kind
V242__poll_wait_purge.sql	DELETE grant and ended_at index for the purge (renumbered from V212)
orqix.polling.tick	Scheduler tick, default 2 s
orqix.polling.retention / purge-delay	Ended-row retention, default 30 days; purge hourly

Main files: `PollTaskDelegate, PollScheduler, PollWaits, WaitStates, MessageWaits, RunRepair, ReleaseCompatibility, InsightsQueries` (runtime); `PollRules, PollSchedule, GraphRules, SimulationRun` (compiler); `PollFields.tsx, FallbackCheckFields.tsx, WaitingForPanel, OutsideWaits.tsx, pollSchedule.ts` (frontend); `contracts/workflow-v0.schema.json`.

4. Connection definition v1 and lookup reconciliation (S130)

A new sibling schema `contracts/connection-definition-v1.schema.json` (schemaVersion 1.0, kind http). Version 0.1 definitions are read, validated, compiled and executed exactly as before; the schema is chosen by schemaVersion. Workflows still map flat inputs; the wire shape is declared in the connection.

Addition	Contract	Safety rule
Path templates	/api/now/table/incident/{sysId} with typed pathParams (pattern, enum or const)	Percent-encoded; empty, '.', '..', '/', '\' and control characters refused; text needs a pattern or enum
Query	const, input, or template with encoder rfc3986 or servicenowEncodedQuery	Encoders are reviewed code; the ServiceNow encoder refuses ^ and line breaks in values
Headers	Static or mapped headers; responseHeaders readable	Authorization, Cookie, Host, Content-*, hop-by-hop, Proxy-*, Sec-*, X-Orqix-* refused
DELETE	method DELETE	Cannot be readOnly
Bodies	Templates; json, form, multipart, binary	Depth 5, 200 leaves; files only from this request's attachments, not quarantined, at most 10 MiB
Results	Array indexes, /[*] row lists, /@headers/Name	At most 100 items per list; depth 5
Errors	errors: [{status, code, bodyPointer?, message?}]	Mapped 4xx becomes a known rejection with failureCode; 401/403/408/429 keep platform meaning; 5xx cannot be mapped
Pagination	offset, cursor, pageToken, linkHeader with maxPages <= 10, maxItems <= 100	GET and readOnly only; next links must stay under the binding base URL
Timeouts	maxTimeoutSeconds up to 120 with long: true; workflow timeoutSeconds max raised to 120	Job lock re-extended when needed
Async, rate limit	Declared and validated	Execution comes with CG-W02 (next batch)

Lookup reconciliation (idempotentByLookup)

For systems without idempotency keys that store an external reference (ServiceNow `correlation_id`). Orqix writes a stable key (request reference, step and occurrence, for example `ORQ-1042-raise-1`) on every send. When the outcome is uncertain (timeout, reset, 408, 5xx, or a crashed attempt with an expired lease) it runs the read-only lookup: **found** records success with the found record and sends nothing again; **not found** allows a resend with the same key; a failed lookup leaves the outcome unknown as before. The lookup is retried 3 times over `lookupSettleSeconds` (default 30). `idempotentNatural` (PUT, PATCH, DELETE) retries like idempotentWithKey without a header; appending operations stay nonIdempotent.

Database: `V230__connection_v1_effect_classes.sql` widens the effect_journal effect-class check. Rule codes `CONN_V1_*` live in `ConnectionContracts.java`; `WF_MAPPING_RECONCILE_KEY` stops workflows mapping the key field.

Main files: `ConnectionContracts, V1Requests, V1Results, HttpExecutor, EffectDispatcher, EffectJournal, HttpTaskDelegate, RequestFiles, OpenApiConverter, ConnectionAdminService`; frontend `connectionSchema.ts, HttpInspector.tsx`; example `examples/connections/servicenow-table.connection.json` with WireMock fixtures.

5. Step retry policy, Retry-After, circuit breaker and connection health (S131)

```
"retry": { "maxAttempts": 6, "delay": "PT10S", "multiplier": 3, "maxDelay": "PT30M", "honorRetryAfter": true }
```

- Allowed on http, externalTask and sendAdapter steps. Without it a step keeps execution.maxAttempts and retryDelaySeconds and its BPMN is unchanged.
- Bounds: maxAttempts 1–10, multiplier 1–4, delay 5 s to 1 h, maxDelay at most 6 h. Rules `WF_RETRY_DELAY_INVALID, _TOO_SHORT, _TOO_LONG, WF_RETRY_MAX_DELAY_TOO_LONG, _BELOW_DELAY`.
- Attempt n waits min(maxDelay, delay × multiplier^(n–1)) ± 10 %, computed in `OrqixFailedJobCommandFactory` via `StepRetryPolicy`. Retry-After (seconds or HTTP date) of a 429/503 is used when it is not longer than maxDelay.
- Unknown and permanent outcomes still never retry automatically.
- **Circuit breaker per binding** (`connection_circuit`): 10 retryable failures in a row within 5 minutes open it; calls wait 1 minute (doubling to 15) **without using an attempt**; one trial call decides.

- **Request limit per binding**: `PUT /workspaces/{ws}/connections/{key}/rate-limit` (connection.manage), split across live pods.
- **Connection health**: one row per call kept 24 hours (no payloads); hour and day figures, last error, circuit state and notices CIRCUIT_OPEN, CIRCUIT_TRIAL, TOKEN_REFRESH_FAILING, LOW_SUCCESS_RATE, RATE_LIMITED_BY_SYSTEM, CALLS_DEFERRED in Settings → Connections.

Item	Detail
V220__connection_health_and_circuit.sql	connection_call, connection_circuit, connection_rate_limit, integration_pod; external_job retry columns
API	listConnectionHealth, getConnectionHealth (connection.read); putConnectionRateLimit (connection.manage)
Configuration	orqix.integration.circuit.failure-threshold (10), window (5m), cooldown (1m), max-cooldown (15m)
Behaviour change	A burst of 10 retryable failures across requests of one binding now pauses that binding for a minute instead of each request retrying on its own

6. Secret providers and worker business errors (S132)

Secret references

Reference	Store	Enable with
crm-token	Mounted file (Kubernetes/OpenShift Secret), as before	Always on
vault:<mount>/<path>#<key>	HashiCorp Vault KV v2 (token file or Kubernetes auth)	ORQIX_VAULT_ENABLED, ORQIX_VAULT_ADDR, ORQIX_VAULT_AUTH, ORQIX_VAULT_TOKEN_FILE or _K8S_ROLE
aws-sm:<id or ARN>#<key>	AWS Secrets Manager (EKS web identity or environment keys)	ORQIX_AWS_SECRETS_ENABLED, AWS_REGION
azure-kv:<vault>/<secret>[/<version>]	Azure Key Vault (workload or managed identity)	ORQIX_AZURE_KEYVAULT_ENABLED
gcp-sm:<project>/<secret>[/<version>]	Google Secret Manager (workload identity)	ORQIX_GCP_SECRETS_ENABLED

- SPI `com.orqix.integration.secrets.SecretProvider` behind the unchanged `SecretStore` methods. No cloud SDKs: small JDK HTTP clients (AWS Signature V4 included), so the dependency set does not grow.
- Remote values cached 30 s (`ORQIX_SECRETS_CACHE_TTL`, 0s disables); missing values and failures are never cached; store calls time out after 5 s.
- **Secret-access audit**: counts per workspace, reference, store and day (resolved, cache hits, not found, failed). `GET /workspaces/{ws}/secret-access?days=7` (connection.manage). No value, hash or length is ever stored or logged.
- Store references are accepted on connection bindings (and OAuth client-credential JSON secrets); adapters, triggers, Kafka and webhooks still use mounted files.
- Database: `V240__secret_providers_and_access_audit.sql` (secret_ref check, secret_access_count).

Worker business errors

- `POST .../external-jobs/{jobId}/fail` accepts `"errorCode": "SUPPLIER_BLOCKED"` (pattern ^[A-Z][A-Z0-9_]{1,39}$, final; retryable true or ORQIX_ codes give 422).
- The step takes its **Failed** route with outputs failureKind (workerError), failureCode, failureMessage, or with `raiseErrorOnFailure` the section's error route; otherwise the request needs attention as before.
- When the engine stops a worker step (section time limit, another error, request ended) its open job is cancelled and a late completion gets 409.
- Python SDK: `BusinessError(code, message)` or `Worker.fail_with_error(job, code, message)`. Studio: Failed handle and route on worker steps.
- Database: `V241__external_job_business_errors.sql` (external_job.error_code).

7. Plan document

`docs/29-camunda-gap-implementation-plan.md` compares Orqix with Camunda 7 and 8 up to 8.10, lists 16 Camunda drawbacks with the design rule each teaches, specifies waiting patterns, connector packages, AI decisions and MCP tools in detail, and contains the full gap matrix and delivery waves. The README reading path links it.

8. Fixes made while integrating

The four features were built in parallel and merged here. These are the problems the combined build found and how each was resolved. Redo them by hand only when cherry-picking feature commits without the merge commits.

#	Problem	Fix	Where
F1	Retry and connection v1 both changed the call result record (`HttpExecutor.CallResult`) and the failure record (`EffectDispatcher.Failed`)	Records carry all fields (page, failureMessage, retryAfter / failureCode, failureMessage, retryAfter) with compatible constructors; the circuit guard wraps the v1 call path (lookups and pagination included)	Merge `22c7c59b`
F2	Paginated call result built with the old field count; the app did not compile	Pass the new retryAfter field (null) in the paginated result	`9f80e10b`
F3	Poll checks did not handle a deferral from the circuit breaker or rate limit, so a held-back check counted as an error	Scheduler catches the deferral, moves the next check without counting a check or error, gives up at the deadline; Retry-After used for the next check	`49541af6`
F4	Migration V212 sorted below V220–V241; a database at V241 would refuse it	Renumbered to V242 and index created IF NOT EXISTS, safe on a database that ran V212	`1fc2d70b`
F5	Worker business errors and step retry both changed the external-job failure branch	Business error routing first, then retry with the step's computed delay	Merge `b6adb99f`
F6	Independent additions in the same places (simulator, test stubs, route rules, BPMN attributes, admin guide)	Kept both sides; external worker steps get the Failed route next to polling's timedOut	Merges
F7	Generated frontend API types conflicted	Regenerated from the merged OpenAPI and checked	Merges

9. Verification

Suite	Result
Compiler (backend/compiler)	884 tests, 0 failures
Application (backend/app, PostgreSQL and Flowable via Testcontainers, WireMock, Vault container)	1,239 tests, 0 failures, 5 skipped (2 live-AI tests, the opt-in polling load test and existing opt-ins)
Polling load test (run by the waits session)	2,000 concurrent polling waits against WireMock; measured check rate recorded in S129
Frontend (Vitest, 204 files)	1,415 tests pass; typecheck, ESLint and OpenAPI type check pass
Python worker SDK	38 tests pass
Not run	Docker stack, real-browser journeys, live ServiceNow or cloud secret stores, production gates

10. Known limits and what comes next

- Each poll check adds one effect-journal row (up to 1,000 per step occurrence).
- Rate limits are per pod (limit divided by live pods); exact global limits need a shared limiter (decision D-CG8).
- No Studio editor for `config.retry` yet; it is set in the workflow JSON or imported BPMN.
- Not yet in v1 connections: XML/SOAP templates, odataFilter and jql encoders, async execution, Basic/JWT/mTLS bindings, row lists in the OpenAPI import.
- Next batch (proposed): connector packages and the ServiceNow package with start-and-follow-up (rest of Wave A), AI decision steps, agent tools with a durable loop, and an Orqix MCP server (Wave B).

Appendix: files changed per commit

A = added, M = modified, D = deleted, R = renamed. Generated from Git.

b931c3a5 docs: Camunda gap analysis, drawbacks and implementation plan

```
M README.md
A docs/29-camunda-gap-implementation-plan.md
```

1411a52a feat(compiler): Keep checking until done and message-wait safety net (CG-W01, CG-W03)

```
A backend/compiler/src/main/java/com/orqix/compiler/PollSchedule.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/PollRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/PollRulesTest.java
M contracts/workflow-v0.schema.json
A examples/connections/order-status.connection.json
```

b19e7d09 feat(runtime): poll scheduler, safety-net checks and the Waiting for API (CG-W01, CG-W03, CG-W08)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/runtime/api/WaitDtos.java
A backend/app/src/main/java/com/orqix/runtime/api/WaitsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixRoute.java
A backend/app/src/main/java/com/orqix/runtime/engine/PollTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/RuntimeProjectionListener.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageDelivery.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/OperatorService.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollConditions.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollScheduler.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunStore.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitStates.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitsService.java
A backend/app/src/main/resources/db/migration/V210__poll_wait.sql
A backend/app/src/main/resources/db/migration/V211__wait_state.sql
A backend/app/src/test/java/com/orqix/runtime/PollWaitsTest.java
M backend/app/src/test/java/com/orqix/support/Stub.java
A backend/app/src/test/java/com/orqix/support/TrackingStub.java
M contracts/openapi.yaml
```

c667c1c4 feat(studio): Keep checking until inspector, Waiting for panel, simulator check sequences; S129 note

```
M contracts/openapi.yaml
M docs/build/README.md
A docs/build/S129-check-until-and-waits.md
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/RunDetailPage.tsx
A frontend/src/features/runs/WaitingForPanel.test.tsx
A frontend/src/features/runs/WaitingForPanel.tsx
M frontend/src/features/studio/inspector/HttpInspector.tsx
A frontend/src/features/studio/inspector/PollFields.test.tsx
A frontend/src/features/studio/inspector/PollFields.tsx
A frontend/src/features/studio/model/pollSchedule.test.ts
A frontend/src/features/studio/model/pollSchedule.ts
M frontend/src/features/studio/test/StepSettings.tsx
M frontend/src/features/studio/test/scenarioModel.ts
M frontend/src/mocks/callbacks.ts
A frontend/src/mocks/pollSimulator.test.ts
M frontend/src/mocks/simulator.ts
A frontend/src/mocks/waits.ts
```

78865323 feat(integration): connection definition v1 and lookup reconciliation (CG-C01, CG-C02)

```
M backend/app/src/main/java/com/orqix/catalog/export/WorkspaceExportService.java
M backend/app/src/main/java/com/orqix/integration/ConnectionDefinition.java
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
M backend/app/src/main/java/com/orqix/integration/EffectJournal.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/RequestFiles.java
A backend/app/src/main/java/com/orqix/integration/V1Requests.java
A backend/app/src/main/java/com/orqix/integration/V1Results.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
M backend/app/src/main/java/com/orqix/integration/openapi/OpenApiConverter.java
M backend/app/src/main/java/com/orqix/integration/openapi/OpenApiReview.java
M backend/app/src/main/java/com/orqix/integration/package-info.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunRequestFiles.java
A backend/app/src/main/resources/db/migration/V230__connection_v1_effect_classes.sql
A backend/app/src/test/java/com/orqix/api/ConnectionV1ApiTest.java
M backend/app/src/test/java/com/orqix/api/WorkspaceExitExportApiTest.java
A backend/app/src/test/java/com/orqix/integration/ServiceNowFixturesTest.java
A backend/app/src/test/java/com/orqix/integration/V1RequestBodiesTest.java
M backend/app/src/test/java/com/orqix/integration/openapi/OpenApiConverterTest.java
A backend/app/src/test/java/com/orqix/runtime/ServiceNowConnectionTest.java
M backend/app/src/test/java/com/orqix/support/OpenApiContract.java
A backend/app/src/test/java/com/orqix/support/ServiceNowStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M backend/compiler/build.gradle
M backend/compiler/pom.xml
A backend/compiler/src/main/java/com/orqix/compiler/ConnectionContracts.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Connections.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/schema/SchemaValidation.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/ConnectionV1CasesTest.java
A contracts/connection-definition-v1.schema.json
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
A docs/build/S130-connection-definition-v1.md
M docs/build/S4-workflow-compiler.md
A examples/connections/servicenow-table.connection.json
A examples/fixtures/connection-v1-cases.json
A examples/fixtures/servicenow-table.wiremock.json
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/operatorText.ts
M frontend/src/features/studio/connectionSchema.ts
M frontend/src/features/studio/inspector/HttpInspector.tsx
A frontend/src/features/studio/inspector/HttpInspector.v1.test.tsx
M frontend/src/features/studio/useConnectionLookup.ts
M tools/contract-check/contract_check.py
```

c07156bc feat: step retry policy, Retry-After, circuit breaker and connection health (CG-W04, CG-C09)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/integration/CallDeferredException.java
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/RetryAfter.java
A backend/app/src/main/java/com/orqix/integration/api/ConnectionHealthController.java
A backend/app/src/main/java/com/orqix/integration/health/BindingRateLimiter.java
A backend/app/src/main/java/com/orqix/integration/health/CallGuard.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionCircuit.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionHealth.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionHealthService.java
M backend/app/src/main/java/com/orqix/runtime/engine/EngineCommands.java
M backend/app/src/main/java/com/orqix/runtime/engine/ExternalTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixFailedJobCommandFactory.java
M backend/app/src/main/java/com/orqix/runtime/internal/ExternalJobService.java
M backend/app/src/main/java/com/orqix/runtime/internal/FailedJobRecorder.java
M backend/app/src/main/java/com/orqix/runtime/internal/JobFailure.java
A backend/app/src/main/java/com/orqix/runtime/internal/StepRetryPolicy.java
A backend/app/src/main/resources/db/migration/V220__connection_health_and_circuit.sql
M backend/app/src/test/java/com/orqix/resilience/Nodes.java
A backend/app/src/test/java/com/orqix/runtime/StepRetryAndCircuitTest.java
A backend/app/src/test/java/com/orqix/runtime/internal/StepRetryPolicyTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
```

```
M backend/app/src/test/java/com/orqix/support/RuntimeTestBase.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnDefaults.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/StepRetry.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/RetryRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/test/java/com/orqix/compiler/BpmnSourceTest.java
M backend/compiler/src/test/java/com/orqix/compiler/CompilerCasesTest.java
M backend/compiler/src/test/java/com/orqix/compiler/GoldenFilesTest.java
A backend/compiler/src/test/java/com/orqix/compiler/StepRetryTest.java
M backend/compiler/src/test/resources/golden/new-hire-onboarding/hashes.json
M backend/compiler/src/test/resources/golden/new-hire-onboarding/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/order-payment-wait/hashes.json
M backend/compiler/src/test/resources/golden/order-payment-wait/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/purchase-approval/hashes.json
M backend/compiler/src/test/resources/golden/purchase-approval/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/start-date-onboarding/hashes.json
M backend/compiler/src/test/resources/golden/start-date-onboarding/process.bpmn20.xml
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/admin-guide.md
M docs/build/README.md
A docs/build/S131-retry-and-circuit-breaker.md
M docs/build/S4-workflow-compiler.md
M docs/user-guide.md
M examples/fixtures/compiler-cases.json
M frontend/src/api/schema.d.ts
A frontend/src/features/settings/ConnectionHealthPanel.test.tsx
A frontend/src/features/settings/ConnectionHealthPanel.tsx
M frontend/src/features/settings/ConnectionsPanel.tsx
A frontend/src/features/settings/connectionHealthApi.ts
M frontend/src/lib/events.test.ts
M frontend/src/lib/events.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/connectionHealth.ts
M frontend/src/test/server.ts
```

003ea821 feat: secret providers with access audit and worker business errors (S132)

```
M backend/app/src/main/java/com/orqix/integration/adapters/InboundService.java
M backend/app/src/main/java/com/orqix/integration/api/ConnectionDtos.java
A backend/app/src/main/java/com/orqix/integration/api/SecretAccessController.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
M backend/app/src/main/java/com/orqix/integration/internal/SecretStore.java
M backend/app/src/main/java/com/orqix/integration/kafka/KafkaConsumerRunner.java
A backend/app/src/main/java/com/orqix/integration/secrets/AwsSecretsManagerProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/AwsSigV4.java
A backend/app/src/main/java/com/orqix/integration/secrets/AzureKeyVaultProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/BearerTokens.java
A backend/app/src/main/java/com/orqix/integration/secrets/FileSecretProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/GoogleSecretManagerProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretAccessAudit.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretAccessQueries.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretHttp.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretProviderSettings.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretRef.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretResolver.java
A backend/app/src/main/java/com/orqix/integration/secrets/VaultSecretProvider.java
M backend/app/src/main/java/com/orqix/integration/webhooks/WebhookDispatcher.java
M backend/app/src/main/java/com/orqix/runtime/api/ExternalJobDtos.java
M backend/app/src/main/java/com/orqix/runtime/engine/ExternalTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/internal/ExternalJobService.java
M backend/app/src/main/resources/application.yaml
A backend/app/src/main/resources/db/migration/V240__secret_providers_and_access_audit.sql
A backend/app/src/main/resources/db/migration/V241__external_job_business_errors.sql
A backend/app/src/test/java/com/orqix/integration/secrets/SecretProvidersTest.java
A backend/app/src/test/java/com/orqix/integration/secrets/SecretProvidersVaultTest.java
A backend/app/src/test/java/com/orqix/runtime/ExternalWorkerErrorsTest.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ScopeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/GeneratedXmlTest.java
A backend/compiler/src/test/java/com/orqix/compiler/WorkerBusinessErrorsTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/admin-guide.md
A docs/build/S132-secret-providers-and-worker-errors.md
M frontend/src/api/schema.d.ts
M frontend/src/features/studio/StepNode.tsx
M frontend/src/features/studio/StepsView.tsx
M frontend/src/features/studio/StudioDialogs.tsx
M frontend/src/features/studio/inspector/ExternalTaskInspector.test.tsx
M frontend/src/features/studio/inspector/ExternalTaskInspector.tsx
M frontend/src/features/studio/inspector/ScopeInspectors.tsx
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/ops.ts
M frontend/src/features/studio/notation/BpmnStepNode.tsx
M tools/worker-sdk/README.md
M tools/worker-sdk/orqix_worker.py
M tools/worker-sdk/test_orqix_worker.py
```

9f80e10b fix(integration): pass retryAfter in the paginated call result after the Wave A merge

```
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
```

49541af6 feat(waits): close S129 follow-ups after the Wave A merge

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/insights/api/InsightsController.java
M backend/app/src/main/java/com/orqix/insights/api/InsightsDtos.java
M backend/app/src/main/java/com/orqix/insights/internal/InsightsQueries.java
M backend/app/src/main/java/com/orqix/insights/internal/InsightsService.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/ResultMappings.java
M backend/app/src/main/java/com/orqix/integration/V1Results.java
M backend/app/src/main/java/com/orqix/runtime/internal/OperatorService.java
M backend/app/src/main/java/com/orqix/runtime/internal/PollScheduler.java
M backend/app/src/main/java/com/orqix/runtime/internal/PollWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/ReleaseCompatibility.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunRepair.java
M backend/app/src/main/java/com/orqix/runtime/internal/WaitStates.java
A backend/app/src/main/resources/db/migration/V212__poll_wait_purge.sql
A backend/app/src/test/java/com/orqix/runtime/PollLoadTest.java
M backend/app/src/test/java/com/orqix/runtime/PollWaitsTest.java
M backend/app/src/test/java/com/orqix/support/TrackingStub.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/PollRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/PollRulesTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/build/S129-check-until-and-waits.md
M frontend/src/api/schema.d.ts
M frontend/src/features/insights/InsightsPage.tsx
A frontend/src/features/insights/OutsideWaits.test.tsx
A frontend/src/features/insights/OutsideWaits.tsx
A frontend/src/features/studio/inspector/FallbackCheckFields.test.tsx
A frontend/src/features/studio/inspector/FallbackCheckFields.tsx
M frontend/src/features/studio/inspector/MessageInspector.tsx
M frontend/src/features/studio/inspector/PollFields.tsx
M frontend/src/features/studio/model/ops.ts
```

```
M frontend/src/features/studio/model/pollSchedule.ts
M frontend/src/mocks/simulator.ts
M frontend/src/mocks/waits.ts
```

The final rename (`1fc2d70b`): `backend/app/src/main/resources/db/migration/V212__poll_wait_purge.sql` → `V242__poll_wait_purge.sql`, plus `docs/build/S129-check-until-and-waits.md`.